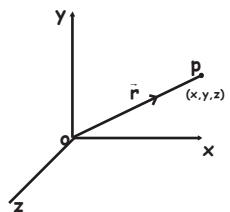


VECTORS

Position Vector

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

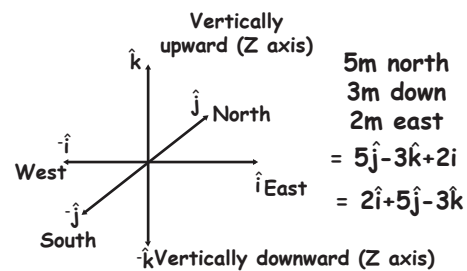
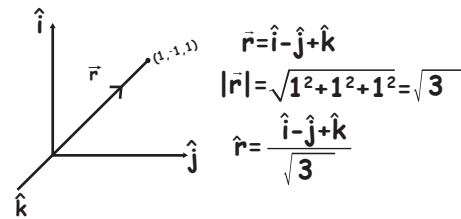
$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$



$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

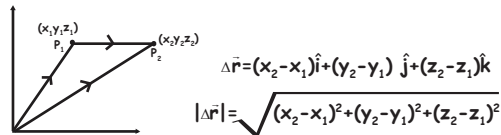
$$|\vec{r}| \text{ or } r = \sqrt{x^2 + y^2 + z^2}$$

$$\text{unit vector} = \hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$$

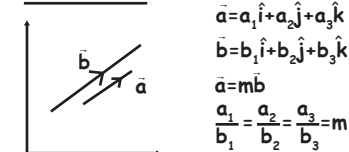


Displacement Vector

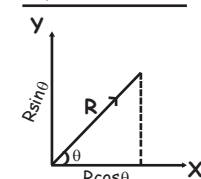
Particle displaces from position P_1 to position P_2



Parallel Vectors

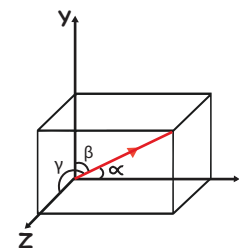


Components of Vector



In two dimensions

$$\vec{R} = R \cos \theta \hat{i} + R \sin \theta \hat{j}$$



In three dimensions

$$\vec{R} = R \cos \alpha \hat{i} + R \cos \beta \hat{j} + R \cos \gamma \hat{k}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

Addition Of Vectors

$$\vec{R} = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$R_{\max} = A + B$$

$$R_{\min} = |A - B|$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

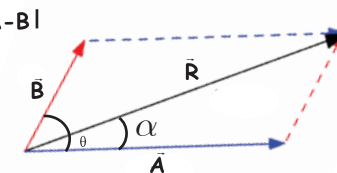
Vector product

$$\vec{C} = \vec{A} \times \vec{B} \quad |\vec{A} \times \vec{B}| = AB \sin \theta$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{i}[A_y B_z - B_y A_z] - \hat{j}[A_x B_z - A_z B_x] + \hat{k}[A_x B_y - A_y B_x]$$

Dot product

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$



PROJECTILE MOTION

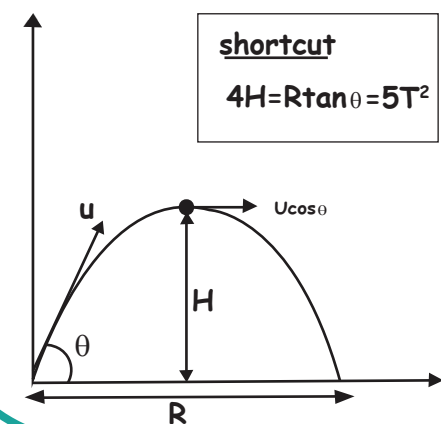
Projectile motion

Horizontal component = $U \cos \theta$

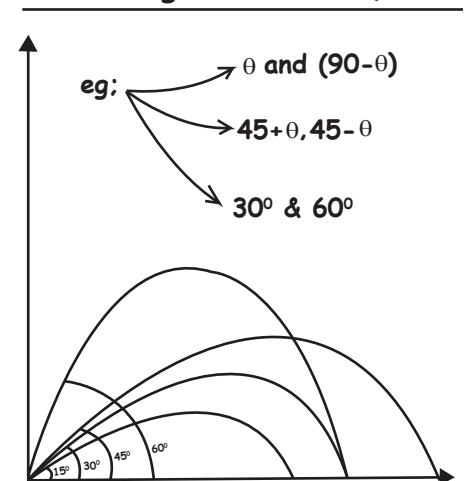
Vertical component = $U \sin \theta$

$$H = \frac{U^2 \sin^2 \theta}{2g} = \frac{(U \sin \theta)^2}{2g} = \frac{U_y^2}{2g}$$

$$R = \frac{U^2 \sin 2\theta}{g} = \frac{2U \sin \theta U \cos \theta}{g} = \frac{2U_x U_y}{g}$$



Same range for θ and $(90 - \theta)$

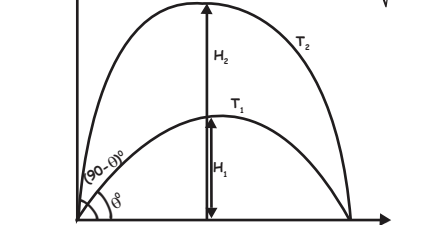


Maximum range

For $\theta = 45^\circ$

$$R_{\max} = \frac{U^2}{g}$$

$$R = 4 \sqrt{H_1 \cdot H_2}$$



$$\frac{T_1}{T_2} = \tan \theta \quad \frac{H_1}{H_2} = \tan^2 \theta$$

$$T_1 \times T_2 = \frac{2R}{g} \quad H_1 \times H_2 = \frac{R^2}{16}$$

$$T_1^2 + T_2^2 = \frac{4R_{\max}}{g} \quad H_1 + H_2 = \frac{R_{\max}}{2}$$

From the relation,

$$4H = R \tan \theta = 5T^2$$

$$4H = R_{\max} \tan 45 \Rightarrow H = \frac{R_{\max}}{4}$$

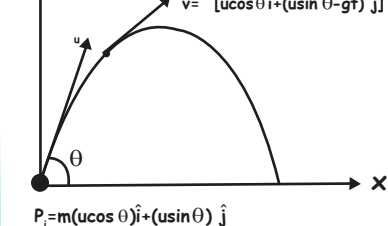
$$4H = R_{\max} \Rightarrow H = \frac{U^2}{4g}$$

$$\text{KE at maximum height} = K \cos^2 \theta$$

Momentum

$$P_f = m(u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j})$$

$$v = [u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}]$$



$$\Delta p = -mgt \hat{j}$$

$$= -mg \times \frac{u \sin \theta}{g} \hat{j}$$

$$= -mu \sin \theta \hat{j}$$

Equation of Velocity

$$V_{\text{net}} = \sqrt{V_x^2 + V_y^2}$$

$$= \sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2}$$

Net displacement

$$x = U \cos \theta t$$

$$y = U \sin \theta t - \frac{1}{2} gt^2$$

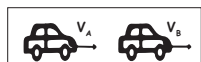
$$r = \sqrt{x^2 + y^2}$$

$$\beta = \tan^{-1}(y/x)$$

RELATIVE MOTION

Relative Motion

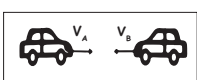
1) Velocity of A with respect to B $V_{AB} = V_A - V_B$



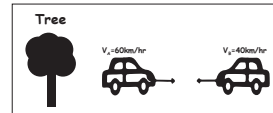
$$3) V_{A/\text{Tree}} = V_A - V_{\text{Tree}} = 60 - 0 = 60$$

$$V_{B/\text{Tree}} = V_B - V_{\text{Tree}} = -40$$

2) $V_{A/B} = V_A - V_B$

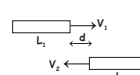


$$= V_A - (-V_B) = V_A + V_B$$

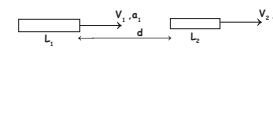


Relative Motion in one dimension (overtaking & chasing)

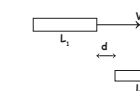
$$1) t = \frac{d + L_1 + L_2}{V_1 + V_2}$$



$$3) d + L_1 + L_2 = (u_1 - u_2)t + \frac{1}{2}(a_1 - a_2)t^2$$



$$2) t = \frac{d + L_1 + L_2}{V_1 - V_2}$$



Minimum separation to avoid collision

$$u_{\text{rel}} = u_1 - u_2$$

$$a_{\text{rel}} = -a_1$$

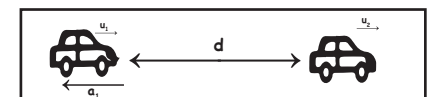
$$0 = u_{\text{rel}}^2 + 2a_{\text{rel}}s_{\text{rel}}$$

$$0 = (u_1 - u_2)^2 - 2a_1s$$

to avoid collision,

$$s \leq d \Rightarrow d \geq \frac{(u_1 - u_2)^2}{2a_1}$$

$$S = \frac{(u_1 - u_2)^2}{2a_1}$$



a_1 = retardation of car 1



RELATIVE MOTION

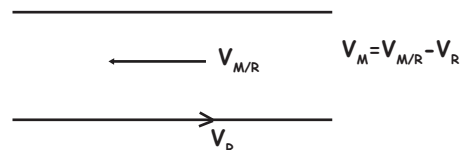
Man-river problem

1) $\vec{V}_{MR}$ or $\vec{V}_{M/Still\ water}$ = velocity due to effort of man, OR velocity of man in still water

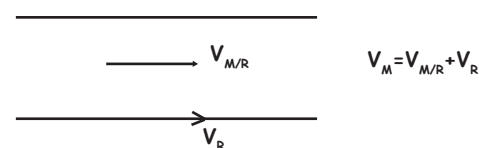
2) V_R = velocity of River

3) $\vec{V}_m$ = Resultant velocity of man with respect to ground

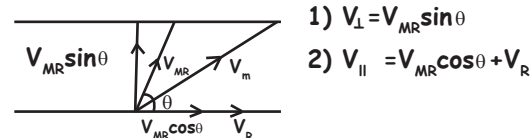
1) Upstream



2) Down stream



Swimming across the river



$$1) V_{\perp} = V_{MR} \sin \theta$$

$$2) V_{\parallel} = V_{MR} \cos \theta + V_R$$

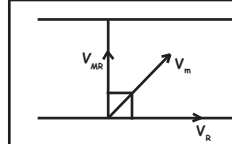
$$t_{cross} = \frac{d}{V_{MR} \cos \theta} = \frac{d}{V_{\perp}}$$

$$X_{drift} = (V_{MR} \sin \theta + V_R) \times t$$

$= V_{MR} \cos \theta \times t$
Due to effort of man

$+ V_R t$
Additional due to push of river

Shortest time



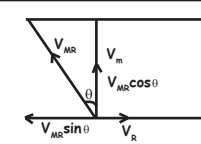
$$\bullet t = \frac{d}{V_{MR} \sin \theta}$$

$$\bullet t_{min} = \frac{d}{V_{MR}}$$

$$\bullet X_{drift} = V_R \times t$$

$$\bullet V_m = \sqrt{(V_{MR})^2 - (V_R)^2}$$

Shortest Path (No drift)



Horizontal component of V_m becomes 0
 $\Rightarrow V_{MR} \sin \theta = V_R$ $V_{MR} \cos \theta = V_m$

Condition for no drifting
 $\Rightarrow \sin \theta = \frac{V_R}{V_{MR}}$

$$\Rightarrow t_{cross} = \frac{d}{V_m}$$

$$\Rightarrow V_m = \sqrt{V_{MR}^2 - V_R^2}$$

$$\Rightarrow \text{Drift} = 0$$

Escalator

$$t_3 = \frac{d}{V_E + V_{M/E}} = \frac{d}{\frac{d}{t_2} + \frac{d}{t_1}} = \frac{t_1 t_2}{t_1 + t_2}$$

t_1 = Time taken by a man to move distance d on a stationary escalator

t_2 = Time taken by a stationary man to move distance d along with moving escalator

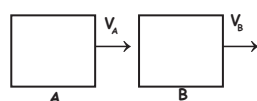
t_3 = Time taken by a man to move distance d while walking along a moving escalator

V_E = Velocity of escalator

$V_{M/E}$ = Velocity of man w.r.t escalator

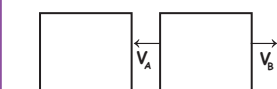
MAN RAIN PROBLEM

Man-rain problem



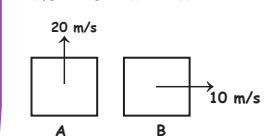
In order to find the relative velocity of B with respect to A we have to reverse the direction of vector A and add it with vector B

w.r.t. A



$$V_{B/A} = V_B - V_A, \quad V_B \text{ w.r.t } A$$

$$V_{A/B} = V_A - V_B, \quad V_A \text{ w.r.t } B$$



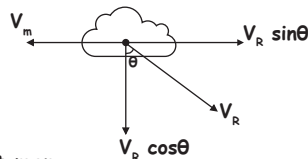
Terms

$V_R \Rightarrow$ Velocity of rain w.r.t stationary man

$V_m \Rightarrow$ Velocity of man

$V_{R/m} \Rightarrow$ Velocity of Rain w.r.t man

Method



$$V_{R/m} = \sqrt{(V_R \sin \theta - V_m)^2 + (V_R \cos \theta)^2}$$

Case 1

$$V_R \sin \theta > V_m$$

$$V_{R/m} = \sqrt{(V_R \cos \theta)^2 + (V_R \sin \theta - V_m)^2}$$

$$\tan \alpha = \frac{V_R \sin \theta - V_m}{V_R \cos \theta}$$

$$\alpha = \tan^{-1} \left(\frac{V_R \sin \theta - V_m}{V_R \cos \theta} \right)$$

Case 2

$$V_m > V_R \sin \theta$$

$$V_{R/m} = \sqrt{(V_R \cos \theta)^2 + (V_m - V_R \sin \theta)^2}$$

$$\alpha = \tan^{-1} \left(\frac{V_m - V_R \sin \theta}{V_R \cos \theta} \right)$$

Case 3

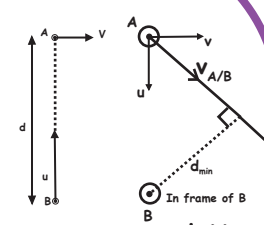
$$V_m = V_R \sin \theta$$

$$V_{R/m} = V_R \sin \theta$$

$$V_{R/m} = V_R \sin \theta$$

$$\alpha = 0$$

minimum distance



$$d_{min} = \frac{d \times V}{\sqrt{V^2 + U^2}}$$

$$\text{Time taken to reach minimum distance} = \frac{d \times U}{V^2 + U^2}$$

CIRCULAR MOTION

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

Angular velocity

$$\omega = \frac{\Delta \theta}{\Delta t} = \frac{2\pi}{T} \text{ (in uniform circular motion), } V = \omega r$$

$$\vec{V} = \vec{\omega} \times \vec{r}$$

Angular acceleration

$$\alpha = \frac{d\omega}{dt} \quad a_t = r\alpha$$

Equation of angular motion

1) Constant angular velocity : $\omega = \text{constant}$

2) Constant angular acceleration

$$\Rightarrow \omega = \omega_0 + \alpha t$$

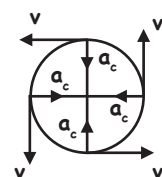
$$\Rightarrow \Delta \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\Rightarrow \omega^2 = \omega_0^2 + 2\alpha(\Delta \theta)$$

Centripetal acceleration

Directed towards centre

Not a constant vector



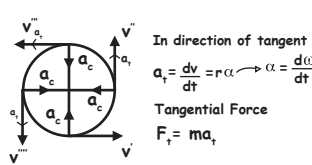
a_c is due to change in the direction of velocity

$$a_c = \frac{v^2}{R} = a_c = r\omega^2$$

$$\vec{a}_c \perp \vec{v}$$

$$\vec{F}_c \perp \vec{s}$$

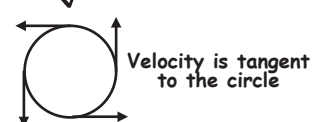
Tangential acceleration



a_t is due to change in magnitude of velocity

Resultant acceleration

$$a_r = \sqrt{a_c^2 + a_t^2}$$



Circular Motion

Uniform Circular Motion

- Speed Constant
- Direction of velocity changes

- $\omega = \text{Constant}$

$$- a_c = \frac{v^2}{r} = r\omega^2$$

$$- a_t = 0$$

Non-uniform Circular Motion

- Speed not Constant

- Velocity changes in direction and magnitude

- a_c = Centripetal acceleration

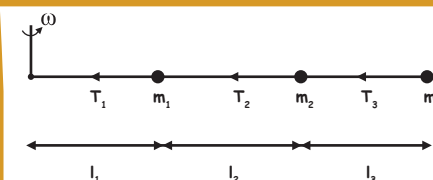
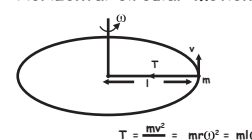
- a_t = tangential acceleration

$$= \frac{dv}{dt}$$

$$- \alpha = \frac{d\omega}{dt}$$

- ω = Changes $\rightarrow \alpha$ angular acceleration

Horizontal circular motion

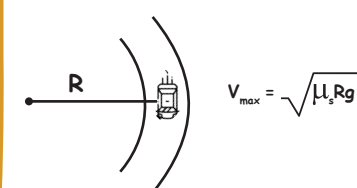


$$T_1 = m_1 l_1 \omega^2 + m_2 (l_1 + l_2) \omega^2 + m_3 (l_1 + l_2 + l_3) \omega^2$$

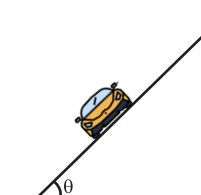
$$T_2 = m_2 (l_1 + l_2) \omega^2 + m_3 (l_1 + l_2 + l_3) \omega^2$$

$$T_3 = m_3 (l_1 + l_2 + l_3) \omega^2$$

Flat circular track



Banking of Road



$$V_{max} = \sqrt{\frac{rg(\tan \theta + \mu)}{1 - \mu \tan \theta}}$$

$$V_{opt} = \sqrt{rg \tan \theta} \quad (\text{for smooth road})$$

$$V_{min} = \sqrt{\frac{rg(\tan \theta - \mu)}{1 + \mu \tan \theta}}$$

At Bottom

$$a) T_{max} = \frac{mv^2}{r} + mg$$

b) min velocity at bottom to complete the circle $= \sqrt{5gR}$

At Top

$$a) T_{min} = \frac{mv^2}{r} - mg$$

b) min velocity at top to complete the circle $= \sqrt{gR}$